

Induwara Gayashan Morawakgoda

Electronic and Telecommunication Engineering Final Year Undergraduate,
University of Moratuwa

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EDUCATION

University of Moratuwa

B.Sc. Engineering (Hons.) in Electronic and Telecommunication Engineering

GPA - 3.73/4.0

- *Deep Learning for Vision, Pattern Recognition, Neural Networks and Fuzzy Logic, IoT Data Structures and Algorithms, Modular Software Development, Software Design Competition*

Moratuwa, Sri Lanka

May 2022 - Present

Sivali Central College

GCE Advanced Level - Physical Science Stream; Z-Score 2.3877

- *Combined Mathematics, Physics, Chemistry*

Ratnapura, Sri Lanka

Jan 2018 - Nov 2020

PROFESSIONAL EXPERIENCE

Software Engineer Intern - WSO2 (Pvt) Ltd

Dec 2024 – May 2025

• **Ballerina HubSpot CRM Commerce Carts Connector (Onboarding Project)** [gh](#)

- * Implemented a Ballerina connector to interact with the HubSpot CRM Commerce Carts API using the OpenAPI specification.
- * Tech Stack: Ballerina.

• **ETL Support for WSO2 Integrator: BI (Main Project)** [gh](#)

- * Developed the Ballerina ETL package, providing APIs for data categorization, cleaning, enrichment, filtering, security, and unstructured data extraction, enabling drag-and-drop ETL workflow creation for Ballerina Integrator (BI).
- * Tech Stack: Ballerina, Java, OpenAI.

GROUP PROJECTS

NeoCare - Contactless Neonatal Health Monitoring System (Final Year Project) [gh](#)

Ongoing

- Contactless neonatal monitoring system estimating heart rate, blood oxygen level (SpO₂), and jaundice status using remote photoplethysmography and deep learning, with on-device inference, privacy-preserving video processing, and mobile app integration.
- Contribution: Implemented a privacy-preserving, motion-aware preprocessing pipeline with ROI extraction, achieving state-of-the-art accuracy for neonatal heart rate estimation.
- Tech Stack: Python, PyTorch, OpenCV, YOLO11, PhysNet, MobileNet.

PowerLens - Transformer Maintenance Record Keeper [gh](#)

Jul 2025 - Nov 2025

- AI-assisted transformer inspection platform for thermal anomaly detection, featuring automated digital maintenance reporting, developed as part of the Software Design Competition module in Semester 7.
- Contribution: Designed the database and backend, implemented bounding box editing, and activity logging.
- Tech Stack: Python, YOLO11, TypeScript, React, Java, Spring Boot, MySQL.

RFID Based Inventory Management System [gh](#)

Nov 2024 - Dec 2024

- RFID-based inventory management system for automated tracking and real-time inventory updates, developed as part of the IoT module in Semester 5.
- Contribution: Implemented the backend and connected it with the RFID system via MQTT.
- Tech Stack: JavaScript, React, Java, Spring Boot, MySQL.

ClariFi - Medical Report/Scan Analyzing Assistant AI [gh](#)

July 2024

- AI-powered assistant for analyzing medical reports and scans, providing automated suggestions to users.
- Contribution: Implemented the OCR feature to extract text from scanned reports.
- Tech Stack: JavaScript, React Native, Python, Django, LLaMA, Firebase.

Vision Based Label Placement Verification System 🔗

Jan 2024 - July 2024

- Vision-based system to detect misprinted labels on a production line, integrated with electronic and mechanical systems for automatically rejecting faulty products, developed as part of the Electronic Design Realization module in Semester 4.
- Contribution: Implemented the object detection module and integrated it with the hardware system.
- Tech Stack: Python, OpenCV, YOLOv8, C++, SolidWorks.

Enterprise Web Application for PCB Manufacturing Operations 🔗

Apr 2024 - May 2024

- A multi-module web application for a PCB manufacturing company, handling orders, inventory, and operations, developed as part of the Modular Software Development module in Semester 4.
- Contribution: Led a team on the customer order management module and contributed to backend development.
- Tech Stack: JavaScript, React, Java, Spring Boot, MongoDB.

Mechapulse - Machine Failure Detection System 🔗

Feb 2024

- A machine failure detection system designed to identify faults in hand drills by analyzing sound variations.
- Contribution: Developed a classification model to detect and classify machine failures.
- Tech Stack: Python, FastAPI

INDIVIDUAL PROJECTS

CodeVision AI - Intelligent Source Code Analyzer 🔗

Jul 2025

- Developed an AI-powered web application that enables intelligent analysis and exploration of GitHub repositories through natural language conversations.
- Tech Stack: Python, Flask, OpenAI, LangChain, Bootstrap, Chroma

MediBot - AI Medical Assistant 🔗

Jul 2025

- Developed an AI-powered medical chatbot that provides accurate health information using RAG technology and a curated medical knowledge base.
- Tech Stack: Python, Flask, OpenAI, LangChain, Bootstrap, Pinecone

Real State Booking System 🔗

Feb 2025 - Jul 2025

- Developed a Real Estate Booking System using the MERN stack, integrating a chat feature for seamless user interaction.
- Tech Stack: JavaScript, React, Node.js, MongoDB

IntelliPrep - Q&A Creator 🔗

Jun 2025

- Developed a web application that generates interview-style Q&A pairs from uploaded PDF documents using OpenAI models.
- Tech Stack: Python, FastAPI, OpenAI, LangChain, JavaScript, React, FAISS

Financial Stock Assistant 🔗

Jun 2025

- Developed an AI-powered stock analysis tool using RAG technology to provide data-driven financial insights.
- Tech Stack: Python, Streamlit, LangChain, OpenAI, ChromaDB

Movie Recommendation System 🔗

Aug 2024

- Developed a movie recommendation system using collaborative filtering to suggest movies based on user preferences and movie similarities.
- Tech Stack: Python, Flask, TypeScript, React

COURSES AND CERTIFICATIONS

Coursera

Machine Learning Specialization(DeepLearning.AI) 🔗

Online

Dec 2023 - Feb 2024

KodeKloud

Docker Training Course 🔗 , Kubernetes Training Course 🔗

Online

Feb 2025

OpenCV

OpenCV Bootcamp 🔗 , TensorFlow-Keras Bootcamp 🔗 , PyTorch Bootcamp 🔗

Online

Jun 2025

ACHIEVEMENTS AND AWARDS

Data Crunch at CodeJam CSE 2025 Finalists Reached the finals of the Data Crunch ML Hackathon (Top 10).	<i>May 2025</i>
Algothan 2024 Finalists Reached the finals of Algothan 2024, a national-level algorithmic programming competition (Top 30).	<i>Dec 2024</i>
Comfix TechCom Ideathon Finalists Reached the finals of the Comfix TechCom Ideathon (Top 10).	<i>Aug 2024</i>
SLIoT Challenge 2023 Semifinalists Developed a machine failure detection system and reached the semifinals (Top 20).	<i>Feb 2024</i>
Idealize 2024 2nd Round Developed <i>ClariFi</i>, an AI-powered assistant for analyzing medical reports and scans.	<i>July 2024</i>
Dean's List Semester 2 (SGPA: 3.84/4.00), Semester 4 (3.96/4.00), Semester 6 (4.00/4.00)	

SKILLS SUMMARY

Languages: English (professional proficiency), Sinhala (native proficiency)
Programming Languages: Python, Java, Ballerina, JavaScript, Matlab
Soft skills: Project Management, Problem Solving, Adaptability

VOLUNTEERING EXPERIENCE

IEEE Robotics and Automation Society - University of Moratuwa	
<i>Co-Lead - Finance Committee</i>	<i>Jan 2025 - Present</i>
<i>Member - Finance Committee</i>	<i>Dec 2023 - Dec 2024</i>
Electronic Club - University of Moratuwa	
<i>External Relationship Coordinator - Main Branch</i>	<i>Oct 2023 - Sep 2024</i>

REFERENCES

Dr. Sampath K. Perera Senior Lecturer Department of Electronic and Telecommunication University of Moratuwa, Sri Lanka Email: sampathk@uom.lk Mobile: +94 70 572 6264	Dr. Kasun T. Hemachandra Senior Lecturer Director, UoM-Dialog 5G Innovation Centre Department of Electronic and Telecommunication University of Moratuwa, Sri Lanka Email: kasunh@uom.lk Mobile: +94 71 895 7734
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